

OGBONNA CHINWEIKE CHIMDIYA

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CAREER SUMMARY

I am an AI, Cybersecurity, and Applied Research professional pursuing a Master's degree, with hands-on experience across machine learning, natural language processing, vulnerability intelligence, and full-stack development. I design and build intelligent, secure systems, from VulnIntel, a dual-engine vulnerability intelligence platform analysing 80,000+ real-world CVE records, to published academic research and human-centric AI applications such as mental health analytics and market prediction. As a teaching assistant in AI-focused labs, I combine research rigour, security-minded engineering, and practical development skills to build solutions that scale and deliver measurable results.

EXPERTISE

Cybersecurity & Vulnerability Intelligence | Threat Detection & Clustering | Artificial Intelligence & Machine Learning | Natural Language Processing | Python Programming | Django Framework & REST APIs | Web Development | Data Visualization | Research & Academic Writing | Data Preprocessing & Feature Engineering | Teaching and Mentorship

EDUCATION

MASTER'S DEGREE | Computer Engineering (M.Tech) | Marwadi University – Gujarat, India 2025 – present

BACHELOR'S DEGREE | Computer Engineering – AI (B.Tech) | Marwadi University – Gujarat, India 2021 – 2025

TECHNICAL SKILLS

LANGUAGES: Python | C programming | C++ | Java | JavaScript | PHP

WEB TECHNOLOGIES: HTML | CSS

AI/ML TOOLS: TensorFlow | Keras | Scikit-learn | NLTK | SpaCy

SECURITY TOOLS: Nmap | OWASP ZAP | Metasploit

LIBRARIES & DATA TOOLS: Pandas | NumPy | Matplotlib | Seaborn

DATABASES: SQL | MySQL | PostgreSQL

PLATFORMS & TOOLS: Git | Linux | VS Code | Jupyter Notebook

PROFESSIONAL EXPERIENCE

TEACHING ASSISTANT (TA) | Marwadi University | Rajkot, Gujarat, India Jul 2025 – present

- Guide undergraduate students through practicals in AI, ML, and programming labs.
- Assist faculty in lab setup, student project reviews, and grading assignments.
- Provide technical support for hands-on sessions in Python, TensorFlow, and NLP frameworks.

FREELANCE WEB DEVELOPER & MANAGER | Eden Associates | Tanzania Jun 2024 – May 2025

- Designed and delivered a full-stack website for a consultancy firm, boosting client interaction by 30%.
- Integrated blog, navigation, contact forms, and responsive design.

PROJECTS & INTERNSHIPS

VULNINTEL: DUAL-ENGINE VULNERABILITY INTELLIGENCE SYSTEM Mar 2026 – present

MTech Dissertation Project

- Designed and built a dual-engine system combining a Discovery Engine (Autoencoder + HDBSCAN) and Guardian Engine (Autoencoder + GMM) for vulnerability clustering and risk scoring.
- Trained and evaluated on a filtered dataset of ~82,000 NVD CVE records (2023–2025), producing 8 distinct threat clusters with defensible silhouette scores (~0.27).
- Built a live, fully async Django dashboard integrating the NVD API, with CVE lookup, cluster visualization, and automated report export.

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING INTERN | TEACHNOOK | India Jun 2023 – Jul 2023

- Built and evaluated supervised ML models for prediction tasks using Scikit-learn.
- Practiced model validation, feature engineering, and cross-validation.

MENTAL HEALTH ANALYSIS AND AI INTERVENTION Dec 2024 – May 2025

- Developed an intelligent model to assess and recommend mental health interventions from user-generated text.
- Combined emotion classification, sentiment tracking, and custom thresholds for alert generation.

CRYPTOCURRENCY MARKET SENTIMENT ANALYSIS

Jul 2024 – Dec 2024

- Analysed Twitter sentiment using NLP to predict crypto trends.
- Achieved an F1-score of 0.89 using logistic regression and custom emotion encoders.

HOUSE PRICE PREDICTION USING DEEP LEARNING

Jul 2024 – Dec 2024

- Developed a deep learning regression model using Keras.
- Achieved low error rate on test data after hyperparameter tuning.

CHATBOT SYSTEM USING PYTHON & DJANGO

Jan 2024 – May 2024

- Built a Django-powered chatbot with session tracking and multilingual support via GPT.
- Integrated feedback rating and chat history features for continuous improvement.

SENTIMENT ANALYSIS USING NLP

Aug 2023 – Dec 2023

- Used NLTK and Scikit-learn to classify Amazon review sentiments with high accuracy.
- Trained and evaluated models using Bag-of-Words and TF-IDF vectors.

MACHINE LEARNING: DIABETES PREDICTION

Dec 2022 – Apr 2023

- Created a classification model (Random Forest) for diabetes risk prediction.
- Achieved >80% accuracy using health parameters and cleaned datasets.

PUBLICATIONS

BOOK CHAPTER: Explainable Artificial Intelligence (XAI) Techniques for Vulnerability Risk Assessment and Prioritization in Modern Industrial and Cyber-Physical Environments

Accepted

- CRC Press (Taylor & Francis) — in "Smart Industry 4.0: Computational Approaches for Optimization and Automation".

PATENTS

SECURITY MONITORING DEVICE — Autonomous Security Robot | 202421091867

Issued – Dec 20, 2024

- Co-invented an autonomous security robot integrating motion, infrared, and proximity sensors with HD/night-vision cameras and GPS-based pathfinding navigation for adaptive, real-time patrol.
- Designed a machine learning-driven threat analysis pipeline enabling real-time anomaly detection, automated alerting, and incident logging, with reduced false positives compared to static security systems.

CERTIFICATIONS

ARTIFICIAL INTELLIGENCE: HUMAN-COMPUTER INTERACTION METHODOLOGIES | Infosys | 2024

AWS ACADEMY CLOUD FOUNDATION | Amazon Web Services | 2023

MACHINE LEARNING | Coursera | 2023

WEB DEVELOPMENT | Pupil First | 2023

OBJECT-ORIENTED PROGRAMMING (JAVA) | Coursera | 2023

INTRODUCTION TO BASH SHELL SCRIPTING | Coursera | 2023

DATABASE PROGRAMMING WITH SQL | Oracle | 2022

PROGRAMMING FOR EVERYBODY (PYTHON) | Coursera | 2022

OTHER ACTIVITIES

- Technical Lead on multiple AI-focused student research projects
- Facilitated and co-organized AI & ML workshops
- Mentored junior students in developing AI capstone projects